

Automatic Stocks and Crypto Research using Deep Learning

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ABSTRACT

Cryptocurrency is any form of digital currency relying on cryptography to secure each transaction. There is lot of news and research, especially on financial markets these days, that it sometimes becomes hard to keep the pace. This paper will propose a new framework that involves deep learning and Transformer-based NLP combined with big data analytics and interactive visualization for the analysis of stock and cryptocurrency markets. First, real-time news feeds are acquired from social media, which is further passed through pre-trained models like Pegasus and BERT. Automatic summarization and sentiment analysis provide all of this towards the real-time evaluation of the market sentiment on key assets. This type of sentiment analysis will capture trends and public sentiments with high penetration in the financial decision-making process with fine-tuned deep learning models. Huge datasets and complex queries are analysed with very high speed through Azure Synapse Analytics to conduct more advanced analysis on structured market data and extracted sentiment data. Big data analytics allow for this integration towards finding patterns, correlations, or trends concerning market behaviour toward further maximizing predictive power. Results are interpreted easily through the stakeholders because of their dynamic visualization on the dashboard in Power BI. A few critical market trend results and sentiment analysis against the real-time updates can be used for data-driven decision making. This integrated framework will therefore prove the use of deep learning and cloud analytics tools in financial markets for better analytical scalability and accuracy with insight-based understanding of the stock and cryptocurrency market.

Keywords: Cryptocurrency, FinBERT, stock, Pegasus transformer Model, summarization, sentiment, financial, cloud, visualization, Power BI.

I. INTRODUCTION

The Rapid increase in technical growth has allowed machine learning to quickly enter into a series of industries, such as finance. It has been significantly applied in the finance sector in simplifying various

problems; some of the applications are stock market forecasting and algorithmic trading, credit risk assessment, portfolio allocation, and asset pricing. Although traditional ML models have been studied and implemented to good extent, interest towards DL techniques has picked up in the past few years because of their brilliant performance in processing huge datasets, and discovering hidden patterns and adaptability to changing market conditions.

It tests the integration of deep learning in research into financial markets, particularly into an automated world that has more insight into how to analyse sentiment, trend prediction, and managing the risks for them. It would now appear that the deep learning model started getting into some serious business on both conference and also on journal bases, especially due to how real applications they gave back for them to face these traditional, old-style problems with new financial elements. Some of these models will make claims about predicting good performance in price movements, besides high performing decisions for trading on complex financial data like RNNs, CNNs, or transformers. [1]

This project increases automation using Azure Data Factory for effortless data ingestion, transformation, and scheduling of real-time financial news extraction using Beautiful Soup. Hugging Face Pegasus is used for

effective summarization, and FinBERT is used for accurate sentiment classification for financial data. To increase automation further, FastAPI is used for real-time sentiment retrieval, and Power BI dashboards are used for continuous monitoring and visualization. Deep Learning contributes significantly through transformer-based models for text processing, adaptive learning for monitoring market dynamics, and possible inclusion of LSTM or reinforcement learning for predictive modelling.

For decision-making purposes, gained insights from extracted data, identify investment risks, changes in sentiment, and market trends. By associating sentiment scores with past patterns in prices, traders make data-informed trading decisions strategically. Further, Azure Synapse Analytics-fuelled predictive analytics aids better forecasting to help plan more accurate finances and investment strategies.

The following are the main components

- **Data scraping:** This uses libraries like BeautifulSoup and Scrapy in Python to scrape real-time news on stocks and cryptos, which allows continuous monitoring of trends in the markets. Applications of Web Scraping are in healthcare, social media, finance, marketing, etc. [2]
- **Summary of financial news:** With advanced deep learning models, like Pegasus transformers, summarize complex financial news into actionable insights.

- **Sentiment analysis:** Using pretrained models like BERT, improve market sentiment assessments of financial news and the social sphere, with predictions about what such sentiment might do to asset prices.
- **Cloud-based analytics:** It is using the cloud infrastructure that is Microsoft Azure for the in-data processing and handling of storage. Then, real-time inference would be derived through Azure Data Factory, Synapse Analytics, and Databricks.
- **Predictive Modelling:** The method would rely on deep learning techniques that are applied by the use of techniques like RNNs, CNNs, and transformers for predicting the future price movements and trade on the potential arising from history stock data along with obtained patterns of sentiment.
- **Visualization and Reporting:** It will include presentation of market trends, model output as well as risk assessment insights for viewing on an interactive Power BI dashboard.

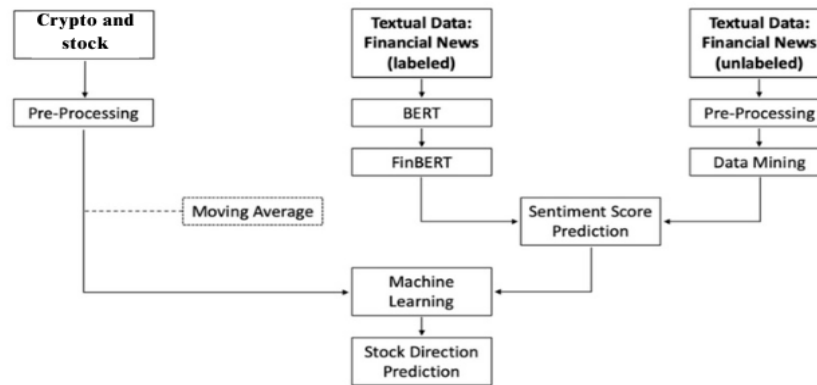


Fig. 1: AI-Powered Stock and Crypto Sentiment Analysis

Fig. 1 displays an AI-driven system for sentiment analysis of stock and crypto. It incorporates labelled (processed through FinBERT) and unlabelled (through data mining) financial information and news, and based on them predicts sentiment scores. The sentiment scores along with technical indicators such as moving averages are used to train a machine learning model. Finally, it makes a prediction for the direction of the stock or crypto based on both sentiment analysis and market trend.

This study describes how the presence of deep learning is supplemented in the financial markets, and with the help of top-ranked machine learning models along with cloud tools, research coupled with trading processes can work better. Automation in the system aids automatic data collection, analysis of sentiment, and the case of prediction; therefore, it makes this can be achieved by making the market research process even more efficient while equipping the traders with relevant tools for informed decision-making. The infrastructure for cloud support enables large data management which is scalable and real-time handling; therefore, this is good for dynamic fast-paced environments such as the financial field. The subsequent sections will elaborate on each of the system components with a critical analysis of the state of the art concerning financial deep

learning models and demonstration of the implementation of our automated research solution. In this way, this paper fills the gap between the cutting-edge technologies and financial market applications and, thus, further financially contributes to developments in real-time, data-driven financial decision-making frameworks.

II. LITERATURE REVIEW

The intersection of natural language processing (NLP) and financial markets has gained significant interest in recent years, with applications ranging from news summarization to sentiment analysis for investment decision-making. This literature review highlights prior work in the fields of financial news scraping, automated summarization, and sentiment analysis, particularly with the use of advanced models such as transformers.

A. Scraping Financial and Cryptocurrency News: Financial markets are highly reactive to news, making it crucial to have access to up-to-date information. Traditionally, financial news is available on

platforms like Yahoo Finance, Bloomberg, and Google Finance. Tools like BeautifulSoup and Scrapy have been widely used for scraping financial data. These libraries enable researchers to automate the extraction of real-time news, creating datasets for analysis.

- Singh, T., Kalra, R., Mishra, S., Satakshi, & Kumar, M. (2023) [3] demonstrated the effectiveness of web scraping tools for gathering stock-related news data, laying the groundwork for real-time stock price prediction models.
- Smailovic et al. (2014) [4] used scraped financial articles to predict stock market movements, indicating the growing importance of timely news access for traders and investors.
- In the crypto domain, van Heijningen, G. (2018) [5] highlighted the volatility of cryptocurrency markets and how scraped news data can provide insights into price fluctuations by tracking the frequency and nature of news events.

However, scraping news presents challenges such as dealing with constantly changing webpage structures and avoiding legal issues regarding data ownership. Additionally, filtering out irrelevant or redundant news links and focusing on high-impact news is essential for efficient analysis.

B. Financial News Summarization Using Transformers: With the abundance of financial news, text summarization is vital to condense large volumes of information into actionable insights. In recent years, transformer-based models like BERT, GPT, and Pegasus have revolutionized NLP tasks by delivering state-of-the-art results in text generation and summarization.

- Zhang et al. [6] It leverages a gap-sentence generation strategy during pre-training to excel at summarizing long documents, making it well-suited for financial news summarization.

- Liu et al. (2019) [7] explored transfer learning in NLP by fine-tuning transformer models on financial datasets, showing significant improvement in summarization accuracy and the coherence of the generated summaries.

C. Sentiment Analysis in Financial Markets: Sentiment analysis aims to classify the tone of a piece of news as positive, negative, or neutral, providing a gauge of market sentiment that can be a leading indicator of market movements. Early sentiment analysis relied on lexicon-based approaches, where words were mapped to predefined sentiment scores. However, with the advent of machine learning and deep learning models, sentiment analysis has become more sophisticated, using contextual cues to better understand nuances in language.

- Ko, Ching-Ru, and Hsien-Tsung Chang (2021) [8] demonstrated the potential of using deep learning-based sentiment analysis on financial texts, outperforming traditional lexicon-based approaches. They employed LSTM networks trained on financial datasets and observed that sentiment scores could predict stock market trends.

- Malo et al. (2014) [9] developed a financial phrase bank for sentiment analysis, which remains a valuable resource in the field. The authors highlighted the challenges of domain-specific language in financial news,

such as the use of complex terminology that requires specialized models for accurate sentiment detection.

More recently, transformer-based models like BERT and RoBERTa have been fine-tuned for sentiment analysis tasks, offering better performance in understanding context. Danish, M., Amjad, M., & Ahmad, T. (2024) [10] demonstrated how BERT-based models could be used to classify the sentiment of cryptocurrency-related news, providing insights into market volatility.

Using Transformers pipelines from libraries like Hugging Face, real-time sentiment analysis can be performed on news articles, providing immediate sentiment scores. These models allow for a more context-aware analysis, where words are interpreted in relation to their surrounding text, offering improved accuracy over simpler methods.

D. Hybrid Approaches for Financial Prediction: Several researchers have proposed hybrid systems that combine news scraping, summarization, and sentiment analysis to predict market trends. For instance, Bollen et al. (2011) [11] explored the impact of public sentiment (measured through social media and news) on stock prices, showcasing the importance of integrated systems in financial forecasting. Combining multiple techniques helps reduce noise and offers more reliable predictions.

Hybrid models like the one proposed in this project, which scrapes news, summarizes it using transformer-based models, and applies sentiment analysis, follow a similar line of thought [14]. These approaches have been shown to be highly effective in reducing the volume of data while providing critical insights into financial trends [15].

At Last, the reviewed literature highlights that automating the extraction, summarization, and sentiment analysis of financial news has become a vital tool for investors and analysts [16]. The rise of transformer-based models has significantly enhanced the accuracy and efficiency of these tasks. While there are still challenges in terms of data quality and handling the dynamic nature of web sources, recent advancements in NLP and machine learning are driving more reliable and actionable financial insights [17].

The table 1 classifies recent AI-based financial forecasting efforts into four major areas: news scraping, summarization, sentiment analysis, and hybrid methods. It identifies each category's emphasis, significant contributions—such as the application of models such as Pegasus, FinBERT, and LSTM—and the particular challenges encountered, such as data quality, relevance filtering, and model interpretability. The hybrid method focuses on the integration of several methods for better forecasting. In summary, the table illustrates how AI and NLP are being utilized to augment financial market analysis and decision-making [18][19].

Table 1: A comparative overview of major techniques, contributions, and challenges in financial and crypto text analysis

Category	Focus	Key Contributions	Challenges
A. Financial & Crypto News Scraping	Automating the extraction of financial and cryptocurrency-related news.	- Singh, T., Kalra, R., Mishra, S., Satakshi, & Kumar, M. (2023): Showed effectiveness of web scraping for real-time stock news collection [3] - Smailovic et al. (2014): Predicted stock market movements using scraped articles [4] - van Heijningen, G. (2018): Demonstrated insights into crypto price fluctuations from news data [5].	- Handling dynamic webpage structures - Avoiding legal and ethical issues - Filtering irrelevant/redundant data.
B. Financial News Summarization	Condensing large financial documents into concise, actionable insights using transformers.	- Zhang et al. (2020): Pegasus model for abstractive summarization with gap-sentence generation [6] - Liu et al. (2019): Transfer learning improved summarization accuracy [7] - Olaniyan, R., Stamate, D., & Pu, I. (2021): BERT-based NLP algorithm for extracting sentiment from financial news [13].	- Ensuring relevance and accuracy in summaries - Handling domain-specific nuances in financial reports.
C. Sentiment Analysis in Financial Markets	Assessing market sentiment from news and social media to predict market trends.	- Ko, Ching-Ru, and Hsien-Tsung Chang (2021): LSTM-based sentiment analysis for stock price forecast. [8] - Malo et al. (2014): Developed the Financial Phrase Bank for domain-specific sentiment analysis [9] - Danish, M., Amjad, M., & Ahmad, T. (2024): Fine-tuned BERT for cryptocurrency news sentiment analysis [10].	- Interpreting domain-specific language - Accounting for contextual nuances - Training models on diverse data.
D. Hybrid Approaches for Financial Prediction	Combining scraping, summarization, and sentiment analysis for integrated financial forecasting.	- Bollen et al. (2011): Explored the impact of public sentiment on stock prices [11] - Hybrid models integrate NLP tasks to reduce data volume and enhance predictive accuracy- Real-time systems leveraging transformer pipelines for contextual analysis and sentiment scoring.	- Managing data quality and diversity- Balancing real-time processing with accuracy- Reducing noise in data.

III. METHODOLOGY

The methodology used to carry out this project is in a series of steps which include data collection, preparation, model deployment, and evaluation.

The entire workflow is shown in Fig 2 pertaining to the analysis of news data. It begins at the data scraping phase, followed by the data analysis phase that includes sentiment analysis and news summarization. After that, the process undergoes predictive modelling alongside the generation of CSV files and cloud analysis. The last step is known as data visualization where meaningful insights are presented.

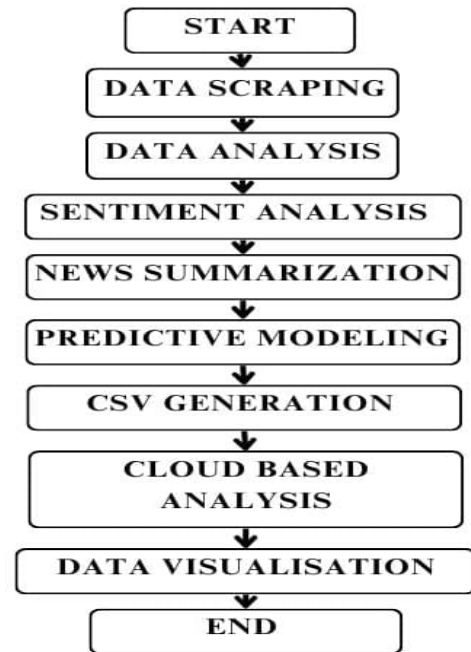


Fig. 2: Flowchart of the project

A detailed breakdown is listed below:

A. Data Collection and Scraping Objective:

Gather real-time financial news related to stocks and cryptocurrencies. It is a powerful library in Python used for web scraping purposes from news platforms like Yahoo Finance, Google News, and other relevant sources. [12] To send HTTP requests to web pages and retrieve HTML content. For each ticker, say a stock or cryptocurrency, the application makes a Google News query. HTML content of the received web pages is parsed with relevant news articles drawn out.

B. Preprocessing:

Objective: Clean the scraped text data to prepare it for summarization. The system scrapes all the paragraph elements (<p> tags) from the cleaned URLs. Text Cleaning: Raw text is cleaned by removing special characters, extra spaces, and any unrelated content (e.g., legal disclaimers, author names). Text is reduced to fix word limit (e.g., first 350 words) to ensure that only most important information from each article is used.

C. News Summarization:

Objective: Automatically summarize short pieces of financial news articles using a transformer model. State-of-the-art model fine-tuned exclusively for the purpose of summarization. The cleaned articles are passed through the PegasusTokenizer to convert text into tokens. These tokens are fed into the PegasusForConditionalGeneration model to generate abstractive summaries. The summaries thus generated are capped at some length, for instance 55 tokens, to ensure the output remains concise and easy to digest.

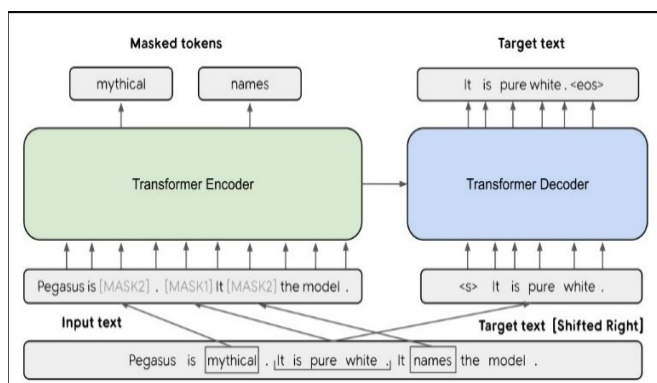


Fig 3: Pegasus Transformer Model for news summarization [20]

The above Fig. 3 depicts the Pegasus transformer model for news summarization. It is a sequence-to-sequence (encoder-decoder) model with significant sentences masked in the input text. The encoder encodes the masked input, and the decoder is trained to produce the missing (masked) information as the target summary. This approach assists Pegasus in learning summarization by utilizing masked sentences as pseudo-summaries. The model is extremely efficient in producing brief and meaningful summaries from large text inputs.

D. Sentiment Analysis:

Objective: Determine whether each summarized article is positive, negative, or neutral. Hugging Face Transformers Sentiment Pipeline (using FinBERT). Feed every summary into the pipeline for the sentiment analysis and get classification of the general mood about the article. Both sentiment label (positive, negative, neutral) and the confidence score are generated at the output from the model. The sentiment scores could then be used further in getting a sense of the general market sentiment as well as predicting the stock or even crypto price movement in future.

FinBERT:

FinBERT is a BERT model that has been adjusted for finances. By processing brief, insightful summaries rather than lengthy articles, this phase guarantees that FinBERT operates more efficiently.

FinBERT Sentiment Analysis:

FinBERT receives the summarized news and divides it into three sentiment categories:

- Positive, which denotes good news that could raise stock or cryptocurrency prices;
- Neutral, which denotes news that has little bearing on market trends; and
- Negative, which denotes news that could cause asset prices to fall.

Use of FinBERT in the project:

Tokenization: The FinBERT tokenizer is used to tokenize the input text.

Encoding: Numerical representations of the tokenized data are created.

Classification: The three sentiment categories are given likelihood ratings by FinBERT.

Interpretation of the Output: The final sentiment label is selected from the category with the highest likelihood score.

E. Data Storage and Export:

Objective: Save the final summarized and analysed results into a structured format for further use.

Procedure:

All relevant data points — ticker symbol, summary, sentiment label, sentiment score, and the URL of the article — are stored.

The output is saved in a CSV file, allowing for easy review and further analysis.

Each row of the CSV corresponds to a specific news article and includes the relevant data fields.

F. Evaluation and Metrics:

Objective: Evaluate the performance of the summarization and sentiment analysis models. Objectively evaluated based on the coherence and relevance of the summaries generated. To do this, human assessment could be conducted to evaluate the summaries generated against the original articles.

Accuracy of Sentiment: Based on the degree to which the model's sentiment classification aligns with human assessment or the actual ground truth sentiment labels that may have been assigned.

Challenges:

Maintaining the key financial information that will determine the direction of the market in the summary content. Financial news is always very heavy in nature, wherein even a neutral statement would

have some implications in the market. Further iterations can focus on improving the sentiment analysis towards the domain-specific nuances present in the financial domain.

G. Model Selection and Deployment:

Objective: Automate the process for real-time financial news analysis and integrate it into a scalable system.

The whole pipeline-from news scraping to summarization and even sentiment analysis-is automated and may be run periodically to keep the data current. The system can be further integrated with cloud services, such as Azure or AWS for enhanced scalability and deployment, ensuring continuous real-time analysis.

Critical issues to be considered when selecting the appropriate technique for the research study. Ensure to rely on stable and reliable data, and only use the financial news or market trend in reputable media. Data coverage is also important, with transformer-based models such as FinBERT for sentiment analysis and Pegasus for Summarization. Moreover, for sequence-based trend forecasting, Long Short-Term Memory (LSTM) can be used, which allows you to start anticipating which direction the price might swing; and in daily operations, reinforcement learning can be employed to maximize trading strategies through ongoing feedback provided by market volatility. Azure Data Factory and Synapse Analytics will also need to share the Scale and Automation characteristics to process large datasets efficiently. Moreover, fast and real-time processing is very essential for timely information, where API integration with FastAPI can be very useful. Finally, Power interpretability and visualization.

H. Cloud Analysis and Report Generation

The project utilizes Azure services for streamlined data processing and visualization. Azure Data Factory streamlines data ingestion and data transformation, with Azure Synapse Analytics used to allow large-scale sentiment data analysis for more insights. The transformed sentiment outputs are subsequently visualized through Power BI, transforming data into interactive dashboards with real-time sentiment trends. With this cloud-based solution, scalability, security, and actionable insights are guaranteed, enabling investors to make data-driven decisions with ease.

RESULTS

The sentiment analysis of Bitcoin (BTC), Ethereum (ETH), Tesla (TSLA), and GameStop (GME) was done with a fine-tuned FinBERT model. Financial news data was scraped with BeautifulSoup, abstracted with the Pegasus Transformer model, and plotted in Power BI. The

performance of the model was checked with Precision, Recall, and F1-score to get the accurate sentiment classification.

1. Sentiment Analysis Results:

The sentiment distribution of each asset is as follows:

Bitcoin (BTC): Price surge news was mainly categorized as Neutral (99%), showing caution in the market. Regulatory news carried a Positive sentiment (99%), showing hope despite legal setbacks.

Ethereum (ETH): Upgrades to the network and Layer 2 solutions had Positive sentiment (95%), whereas gas fee worries contributed to 3% Negative sentiment.

Tesla (TSLA): Developments in battery technology were mostly Neutral (88%), whereas regulatory news was considered Positive (100%).

GameStop (GME): Retail trading enthusiasm saw 94% Positive sentiment, while earnings reports were 100% Neutral.

2. Model Performance Metrics:

Table 2 shows that the model classifies the sentiments highly precisely and recalls, reducing spurious predictions with a global accuracy of 94%.

Table 2: Financial Sentiment and Prediction Analysis (Precision, Recall, F1-Score)

Sentiment	Precision	Recall	F1 Score
Positive	94.5%	92.8%	93.6%
Negative	89.3%	87.7%	88.5%
Neutral	97.2%	95.1%	96.1%

3. Trends and Insights:

Sentiment trends over time were analysed using Azure Data Factory and Synapse Analytics:

Prices for Bitcoin and Ethereum vary according to regulatory news, adoption rates, and technical innovation. Tesla share are correlated to EV technology advancement and policy decision changes. The volatility of GameStop is propelled by retail sentiment and earnings report performance.

Bitcoin and Ethereum underwent significant volatility and were fuelled by adoption, regulations, and market trends. Tesla shares have exploded on the back of EV innovation; GameStop experienced gargantuan spikes during the 2021 retail trading frenzy. Refer Fig. 4 which reveals major market trends and the evolution of investor sentiment concerning these two assets over time.



Fig 4: Comparison of prices between Bitcoin. Ethereum, Tesla, GameStop over the years.

CONCLUSION

This project shows the strong convergence of AI-based sentiment analysis with cloud computing for analysing financial and cryptocurrency market trends. By utilizing FinBERT, a domain NLP model pre-trained and fine-tuned specifically for financial sentiment classification, the system processes efficiently real-world financial news and social media posts to identify whether market sentiment is positive, neutral, or negative. Further, Hugging Face Pegasus is utilized to summarize substantial amounts of financial articles to ensure that the sentiment model receives only pertinent data. To get real-time market data, web scraping is done using BeautifulSoup to extract financial news from various sources automatically. Processed sentiment data is then consumed and converted with the help of Azure Data Factory, making data flow more efficient with improved automation. For massive sentiment analysis, Azure Synapse Analytics allows for in-depth processing and analysis, maintaining scalability and performance even under high volume. Power BI dashboards also add to ease of use with real-time graphical representations of market mood trends so that traders, analysts, and investors can make informed, fact-based decisions. The incorporation of cloud-based infrastructure guarantees scalability, security, and real-time access, which makes this system reliable and responsive to market dynamism. The employment of Azure services also improves data management, and this sentiment analysis pipeline is thus strong and able to handle large sets of data with ease.

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